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Executive Summary

The purpose of this report is to glean useful information from the Fitbit user activity that could influence Bellabeat's marketing and product development strategies. This data includes user engagement with smart devices. I have identified trends in physical activity, sleep habits and other everyday lifestyle behaviours and examined how smart device use can promote wellness and encourage increased engagement with the Bellabeat product.

Generally, the data from Fitbit shows that users are active (an average of 8,515 steps daily and 76.58% of all entries falling under the 'moderately active' and 'highly active' category); however, it also reveals that a significant proportion of the time spent interacting with smart devices includes large blocks of time when the users are being sedentary (11.87 sedentary hours daily). Sleep data indicated that the average user sleeps an average of 6.99 hours per night. This is below the generally recommended quantity of sleep for adults. Analysing weekly trends of activity behaviour discovered a definitive recovery pattern where activity levels spike on Saturday (9,871 steps) while sleep levels are highest on Sunday (7.55 hours). Weekend evenings include an average of 23 more minutes of sleep per night than weekdays, and slightly elevated activity levels. A moderately strong negative correlation between the amount of time spent sedentary and the amount of time spent sleeping was found ($r = -0.61$) and was the most statistically significant finding; as sedentary time increased, sleep time decreased and indicates an opportunity to promote well-being by diminishing lengthy sedentary periods.

Three specific opportunities arise for Bellabeat. First, reducing sedentary time. Second, designing a personalized approach to increased engagement for users with a more sedentary lifestyle. Third, creating campaigns based on users' weekend recovery behaviour to promote consistent well-being by sending reminders, tips, and advertisements.

BUSINESS UNDERSTANDING

Business Task	Analyse smart device fitness data trends in consumer health and fitness behaviour and determine how these insights can be used to improve Bellabeat's marketing strategy.
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Stakeholders:

1- Primary stakeholders:

Urška Sršen	Co-founder & Chief Creative Officer
Sando Mur	Co-founder
Bellabeat marketing Analytics team	Analytics
Bellabeat Executive Leadership team	Executive leadership

2- Secondary Stakeholders:

- Existing Bellabeat customers
- Potential Bellabeat customers
- Smart devices consumers interested in health and wellness product

Business Objective:

Use Fitbit user data to:

- Understand activity habits
- Understand sleep habits
- Identify behavioural trends
- Generate actionable recommendation for Bellabeat

Success Criteria:

The project will be considered successful if:

- Smart devices usage trends are identified
- Trends are supported by data visualization
- Actionable marketing recommendations are provided
- Findings can be presented to Bellabeat executives

DATA SOURCE ASSESSMENT

Dataset used:

Source	Kaggle (made available through Mobius)
License	CC0 Public Domain
Data	collected from 30 Fitbit users
Contains	activity, sleep, heart rate, calories, and step-tracking information
Data period	March 2016-May 2016

Data Credibility Assessment:

- 1- Reliability: The dataset was collected directly from Fitbit devices, reducing manual entry errors.
- 2- Original: The data originated from Fitbit user who consented to share their activity information
- 3- Comprehensiveness: The dataset includes
 - Daily activity
 - Sleep tracking
 - Calories burned
 - Distance travelled
 - Heart rate metrics
- 4- Time: The data was collected in 2016
- 5- Cite: The dataset is publicly available through Kaggle and distributed under a CC0 Public Domain license.

Data description:

In this analysis, the data used is the Fitbit Fitness Tracker Data that forms part of the Bellabeat Case Study. The data entails metrics gathered from Fitbit users, including physical activities, sleeping habits, and sedentary minutes. In this analysis, I have used two main types of data which include daily activity and sleep data.

Upon data validation and data integration, the final data set used in the analysis had 24 different users and 410 days' observations. The variables used in the data include total number of steps, very active minutes, sedentary minutes, number of sleep records, and sleep duration.

Data Limitations

- 1- Small sample size (30 users)
- 2- Data collected only during a short time period
- 3- Data is relatively old (2016)
- 4- No Demographic information available
- 5- Gender information unavailable despite Bellabeat targeting women
- 6- User population may not represent the broader smart-device market

Data Preparation

For data preparation, Python programming language was used to check the quality of the data. Data validation was done by checking for missing values, duplicate records, and incorrect data types. Standardization of date fields was done to allow the merger and comparison of data with respect to time.

The daily activities data and sleep data were merged through unique user ID and activities dates to develop an integrated database that can be analysed later. Several derived variables were developed to help in conducting exploratory analysis. The derived variables included Sleep Hours, Day of Week, Weekend Indicator, and Activity Levels. For activity levels, categories such as Low Active, Moderately Active, and Highly Active were created according to daily steps taken.

After data validation and preparation, data exploration, correlation and dashboards using Tableau were done using the final prepared dataset.

Key Insights

1- Finding 1: 76.6% of all observations are moderately or highly active users, meaning that most Fitbit users in the dataset are physically active.

2- Finding 2: 23.4% of all observations can be considered as low activity users, meaning that there is a group of Fitbit users who might require reminders and activities to increase their level of activity.

3- Finding 3: Users spend 11.87 hours being sedentary on average. The most prominent correlation between a user's sedentary behaviour and other features is found in relation to the duration of sleep ($r = -0.60$).

4- Finding 4: On average, Fitbit users sleep for 6.99 hours per night, which is very close to the recommended 7-hour sleep target.

5- Finding 5: Activity levels remain stable throughout the week both on weekdays (average number of steps 8,480) and during weekends (average number of steps 8,607).

6- Finding 6: The average number of physical activity steps is highest on Saturdays (9,871 average steps) and the average amount of time users spend asleep is highest on Sundays (7.55 hours).

Key Takeaway: Fitbit users generally maintain active lifestyles and achieve near-recommended sleep levels. However, prolonged sedentary behaviour is moderately associated with reduced sleep duration, presenting an opportunity for Bellabeat to promote movement, reduce inactivity, and improve overall wellness engagement.

Recommendation

1- Reduce Sedentary behaviour:

- There was found an inverse association between sedentary behaviour and sleep duration ($r = -0.61$).
- The following recommendations could be made regarding Bellabeat's products: the company should encourage its clients to balance sedentary behaviour using breaks and physical activities through alerts about sedentariness and personal activity goals.

2- Increase Engagement Among Low-Activity Users:

- About 23.41% of the data fall under the low activity category. Bellabeat may use this information to reach out to such users with customized challenges, goal setting options, and motivating notifications.

3- Recommendation 3: Leverage Weekend Recovery Patterns:

- There was a clear pattern that users displayed a high need for recovery during the weekend period, with their sleep durations reaching its peak on Sundays and activities maintaining stability. Bellabeat could benefit from this by offering insights about recovery, sleep, and wellness on weekends.

Conclusion

The analysis of data regarding activity and sleep from Fitbit users identified three important points. First of all, it was found that the majority of users had moderate and high levels of activity, which is considered a positive point for their overall involvement in activities. The second finding is that there is consistency in recovering during the weekend, as evidenced by a longer period of sleep on those days. Moreover, inactivity had the greatest connection with sleep duration – the more inactivity, the less sleep time.

The insights gleaned offer opportunities for Bellabeat to increase wellness through inactive behaviour correction and wellness activities. Through the use of smart devices to understand behaviour, Bellabeat can benefit users with increased wellness and engagement levels.